

1. Constrained Optimization for curriculum composition

After fine-tuning the ClassifyLM model, it can be used to estimate the KA distribution of any given course based on its course description. The model produces a probability distribution over the CSEC2017 Knowledge Areas, providing a representation of the course’s cybersecurity coverage.

The target requirements from the NICE framework for job roles, job categories, and the overall market, are similarly encoded as a probability vector over KAs. This shared vector representation enables the direct alignment of courses with cybersecurity workforce demands, framing the task of optimal course selection as a constrained optimization problem.

To design a curriculum aligned with job market requirements, it is important to consider the structural constraints of the curriculum. In particular, factors such as the total number of credits in the program, and the desired distribution between core and elective credits effectively *constrain* the optimization process. For example, the two-year MSc in Cybersecurity at KTH consists of 120 European Credit Transfer and Accumulation System (ECTS) credits, of which 39.5 ECTS correspond to mandatory core courses, 30 ECTS to conditionally elective courses, 30 ECTS are dedicated to the master thesis project and the remaining 20.5 ECTS are free choice electives. As presented in the case example (in the main body of our paper), CurricuLLM can be used to identify the optimal set of conditional electives for a specific target job role. The program offers 12 elective courses to first-year students, of which students are required to select at least 4. Although CurricuLLM can also be used to recommend free-choice electives, the present scenario focuses on optimizing the selection of conditional electives for simplicity. We formulate the optimization problem for the present scenario as a second-order constrained optimization problem, with the objective of aligning the student’s composed curriculum as closely as possible with the desired KA distribution of the target job role.

Here, we let

- m : Number of KAs, k : Number of elective courses to be selected, n : Total number of available electives, c_M : Total number of mandatory credits.
- c_i the credit attributed to course i .
- $E_{i,j} \in [0, 1]$: Contribution of KA j to elective course i . This is a normalized probability vector, such that $\sum_{j=1}^m E_{i,j} = 1$.
- $T_j \in [0, 1]$: Target KA distribution of the desired job role. This is a normalized probability vector, subject to $\sum_{j=1}^m T_j = 1$.
- $x_i \in \{0, 1\}$: Binary decision variable indicating whether elective i is selected.
- $\mathcal{M} \in \mathbb{R}^m$: The aggregated KA vector of the mandatory courses.

Such that the MIP formulation becomes

$$\text{minimize} \quad \sum_{j=1}^m \left| \frac{c_M}{c_M + \sum_{i=1}^n c_i x_i} \mathcal{M}_j + \frac{1}{c_M + \sum_{i=1}^n c_i x_i} \sum_{i=1}^n c_i x_i E_{i,j} - T_j \right| \quad (1)$$

$$\text{subject to} \quad \sum_{i=1}^n x_i = k \quad (2)$$

$$x_i \in \{0, 1\} \quad \forall i \in \{1, \dots, n\} \quad (3)$$

Each course (mandatory or elective) is represented as a continuous m -dimensional probability vector indicating the proportional contribution of each KA to the course. Specifically, each elective $i \in 1, \dots, n$ is associated with a normalized contribution vector $E_i = [E_{i,1}, \dots, E_{i,m}]^T$. The mandatory curriculum component is similarly represented by a vector $\mathcal{M} \in [0, 1]^m$ and each elective course $i \in \{1, \dots, n\}$ has a KA contribution vector $E_i \in \mathbb{R}^m$. The desired job profile is represented as a target KA distribution $T \in \mathbb{R}^m$, obtained from Table 3 in (Ramezani and Niemi, 2024). Binary variables $x_i \in \{0, 1\}$ are introduced to indicate whether a given elective course i has been selected. The total number of electives selected is constrained to k using constraints (5) and (6). Additionally, because mandatory and elective courses carry different credit values at the program at KTH, we aggregate KA distributions in a credit-weighted manner. We let c_M denote the total number of mandatory credits and c_i the credits of elective i .

The KA distribution of the student’s entire curriculum is then computed as the sum of the mandatory component, $\mathcal{M}$, and the contributions of the selected electives: $\sum_{j=1}^m \sum_{i=1}^n c_i x_i E_{i,j}$, weighted by the number of credits contained in each. The objective (4) is then to minimize the absolute deviation between the KA profile of the

student’s curriculum and the target profile T . This ensures that the selected electives fill the knowledge gap left by the mandatory curriculum as closely as possible.

We next examine to what extent this optimization problem should be reformulated to accommodate alternative curriculum constraints. The one-year MSc in Cybersecurity at Nanyang Technological University (NTU) consists of 30 Academic Units (AU), of which 12 AUs are reserved for mandatory courses and 6 AUs are dedicated towards a capstone project. The remaining 12 AUs are conditionally elective, where students choose four elective courses from a pool of nine. Because of the great structural similarity of this program with the program offered at KTH, CurricuLLM can be applied to the NTU master in a similar fashion as mentioned before.

The two-year M.S. in Information Technology - Information Security (MSIT-IS) offered at Carnegie Mellon University (CMU) consists of 75 core units, 36 free program electives, and 12 restricted electives. While the restricted electives can be chosen in a similar way as posed in the optimization formulation for KTH, the core units have a more complicated structure that we consider outside the scope of our current work. The core units are separated into a networking core (12 units, one course selected out of four options), an MSIT-IS core (12 units, one course selected from three options), a Security core (12 units, fixed), a Business and management component (12 units, comprising either two fixed courses or one alternative course), and a practical project worth 24 units. Additionally, we have not analyzed constraints such as the sequentiality of courses (e.g., Advanced calculus requires prior completion of Basic Calculus), practical constraints such as clashing timeslots, and the student’s prior knowledge.

2. Curriculum Analysis extended results

For each of the 3 curricula analyzed in our paper (KTH, NTU, CMU), we randomly selected 3 courses to be analyzed in further detail. Here, the results for the Building Networked Systems Security (KTH) course are provided earlier in the paper in Table 7. The rest of the results can be found in the following tables: Ethical Hacking (KTH), Table 1, Language-Based Security (KTH), Table 2, Network Security (NTU), Table 3, Software Security (NTU), Table 4, Privacy Preserving Technologies & Security in AI (NTU), Table 5, Advanced Real-World Networks (CMU), Table 6, Networked Systems Course (CMU), Table 7, Information Security Policy and Management (CMU), Table 8.

For each course analyzed, we derived topics from the course descriptions using PreprocessLM, labeled them using ClassifyLM and present the final result of this process under the column CurricuLLM in the table. Then, Control Group A, consisting of 34 cybersecurity experts, and a group of three cybersecurity experts previously familiar with the CSEC2017 framework (noted as ‘Expert Group X’) were consulted to independently label the topics. The numbers listed in the tables down below represent these annotations, where the corresponding KAs for each number can be found in the main body of our paper.

Table 1: Topics extracted from the Ethical Hacking Course (KTH) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Ethical Hacking	1,6	6,8	2,5,6	0–8	2,7,8	0,2–6	8
Network and vulnerability scanning	4	3,4,5,7	1,4	4,5	5	0,4	4
Exploit development platforms	2,7	3	2,3	2	2	2,3,4	2
Command and control	0	2	2,3,6	0	7	0,2,4,5	0
Password cracking	1	5,6	1,2,3	0,1	5	1,2,6	6
Independent attack project	1,3	–	7	0	5	0	5
Virtual environment setup	4	7	1,2	4,5	0	0	4

3. KA distribution NICE Job Roles

The relative importance of each of the nine Knowledge Areas for all work role categories defined in the NICE framework v2.0.0 is shown in the main body of our paper. The ‘relevance’ of KAs was derived by collecting the KDs associated with a given job role, annotating them with corresponding KAs using ClassifyLM, and computing the resulting distribution of KAs. For completeness, this appendix provides the relative importance of the KAs for *all* individual NICE work roles. These detailed results are displayed in Table 9.

Table 2: Topics extracted from the Language-Based Security Course (KTH) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Language-Based Security	1,6	2	–	0,2	6	0,1,2,6,7	5
Fundamental principles, models and concepts for computer security	0	2	1,4,5,7,8	0–5	1,3	0–8	5
Software security by information flow control	3	4	1,2,4	2	2	2	2
Web application and database security	2	2	1,4,5,6,7	0,2	2,5	1,2,4,5,7	7
Security for mobile applications	5,6	2	1,3,4,5,6	4	2,4	2,5,6	7
Hot topics in computer security	1	–	0,5,8	0–8	1,3,4,5	0–8	7
State-of-the-art in programming language for security	1,6	2	2,3,4	2	2	2	7

Table 3: Topics extracted from the Network Security Course (NTU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Network Security	5	3,4,5,7	1,2,5,6,7	4	4,5	4	7
Introduction to Networking	5	1,2,3,4,5	4	0	0	4	4
Security in Networks	5	3,4,5,7	1,2,5,6	4	4,5	4	7
Network Programming	5	4,5	2	0,4	2,4	2,4	7
Packet Trace Analysis with Wireshark	5	4	1,4,7	4	4	3,4	4
TLS and Security in Networked Systems	5	1,2,4,5	1,4,5	4	4,5	4	4

Table 4: Topics extracted from the Software Security Course (NTU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Software Security	3,4,5,7	–	2	2	2	2	2
Challenges in software security	1–5	8	2	2	2	2	2
Principles of secure software development	3–5,7	2	0,2,5	2	2	2	2
Mechanisms for secure coding	4	2,5	5	2	2	2	2
Tools for software security	2,4	1	2	2	2	2	2
Causes of vulnerabilities	1,2,4,5	3,6	0,6,7	0–6	2–5	0–8	7
Strategies to avoid vulnerabilities	1–3	2	3,5,7	0–6	2,3	2,5,6,7,8	7
Defenses against attacks	1–5,7	1,6	3,5,7	0–6	5,7	0–8	5
Secure programming practices	2	3	2,3,5,6,8	2	2	2	2
Language-specific secure coding	2,4	3	2,3,5,7	2	2	2	2
System-level defenses	1,2,4,5	5	1–5	5	5	5	5
Architectural approaches for security	1,3–5,7,8	2,4	0,3–8	4	5	0,5,6,7,8	2
Run-time enforcement techniques	2–6	1,5	0,4,6–8	4	2,3	2,3,5	7
Software analysis tools	1,2,4,6,7	2	0–2,4–8	2	0,2	0,2,3	2
Vulnerability detection methods	1–3,5–8	3	0,2,5–8	3	5	0,2–6	2,3
Application of security tools in scenarios	0,2,5–8	–	0–2,5–8	3	5	0	7

4. Knowledge Area 0: Miscellaneous

In (Ramezani and Niemi, 2024) the authors identified seven knowledge units in the knowledge area 0: Miscellaneous. The description of these knowledge units are as follows.

- *Computer Science*: includes topics related to Programming and Software Development, System Architecture, Database Management, Mathematics, Information Systems, and Human and Organizational Aspects of Information Technologies.

Table 5: Topics extracted from Privacy Preserving Technologies & Security in AI Course (NTU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Privacy Preserving Technologies & Security in AI	1,6	0-3,5,7,8	0-2,6-8	0,2,3,6	5,6,8	0,1,6	1,6-8
Fully Homomorphic Encryption	1	0,1,4	1,3,5	1	1	1	1
Secure Multi-computation	1,2,4	0,2,3,5,7	1,3,5,7	1	1	1	1
Searchable Encryption	1	0,1,4	1,2,5	1	1	1	1
Data Sharing in Machine Learning	1	0,4,7,8	0-2,4,5,7	0,1	0	1,6	4

Table 6: Topics extracted from Advanced Real-World Networks Course (CMU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Advanced Real-World Networks	4,5	1-8	1,4,5	0,4	0	4	4
4G and 5G network infrastructures	4	1-8	0,3-5	0,4	0	1,4,5	4
IPv6	2,5	0	0,1,4	4	0	4	4
SDN and VFN	2	1,2,4,7	0,1,2,3,5	4	0	2,4	4
Data centers	1	1,3-6	0-7	0	0	1-5	1
Mesh networks	3,4	1-5	0,4	4	0	4	4
Embedded networks	3,4	1-8	0,4	4	0	2,4	4

Table 7: Topics extracted from Security in Networked Systems Course (CMU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Security in Networked Systems	1	1-5	1	4	4,5	4	7
Network and transport-layer attacks and defenses	4	1-5	4	4	4,5	4	4
Network intrusion detection	5,6,7	1-5	1,4	4	4,5	4	4
Denial of service (DoS) and distributed denial-of-service (DDoS) detection and reaction	4,5,8	4	1,4	4	4,5	4	4
Worm and virus propagation	2,3	6	2,3,5,7	4	2,4,5	0,2-6	1
Tracing the source of attacks	0-8	0-2,4-6-,8	1,2	0-6	5	0,2,4,5	1
Traffic analysis	4	0,1,4,8	4	4	1,4	0,4	7
Techniques for hiding the source or destination of network traffic	1,2,5	0,1,4,6,8	2,4,5	4	4	4	4
Secure routing protocols	4	0,1,2,4,8	4	4	4	4	4
Content poisoning attacks	1,2,3	1-4	1	4	2-4	2,4	1
Advanced techniques for reacting to network attacks	4,8	0-6	0,6-8	4	4,5	0,4	4

- *Business and Law*: includes topics related to Regulatory Frameworks, Organizational Operations & Policies, Stakeholder Relationships, and Management & Human Resources.
- *Communication and Networking*: includes topics related to Telecommunications Fundamentals, Network Architectures Fundamentals, Network Management, Communication Systems, and Information Gathering and Dissemination.
- *Information Technology*: includes topics related to Technology Infrastructure, Content Systems, Technology Trends & Integration, and Media, Communication, & Dissemination.
- *Cyberspace Practice*: includes topics related to Adversarial Models, Cyber Attacks & Vulnerabilities, Cyber Operations & Practices, Core Cyber Concepts & Principles, and Security Tools, Vendors, & Defensive Technologies.
- *Pedagogy*: includes topics related to Learning Frameworks, Educational Technology, Training Policies,

Table 8: Topics extracted from Information Security Policy and Management Course (CMU) annotated by 3 annotators from Control Group A, all annotators in Expert Group X, and CurricuLLM

Topics	Control Group A			Expert Group X			CurricuLLM
	#1	#2	#3	X ₁	X ₂	X ₃	
Information Security Policy and Management	0,1,5,6,8	7,8	7	7	5,7	0,7,8	7
Security marketplace overview	0,2,8	3	0	7	5,7	0	7
Decision making with multiple parties involved	5,7,8	6,8	8	7	0	0,1,7,8	7,8
Role of policy in information security	1,7	8	7	7,8	5,7,8	7	7,8
Intra-organization policies	4,7	4,8	3,5,6	7	0	7	7
Market and competition impact on security provision	0,7,8	1,2	8	7	5,7,8	0,7,8	7
Causes of market failure (externalities)	0,5,7,8	0,7	6–8	0,7	0	0,7,8	7
Policy tools to mitigate market failure	0,5	0,8	3,7,8	7	0	0,7	7,8
Key laws and regulations on product liability and security standards	5	3,7,8	1,2	7,8	8	0,3,4,5,7,8	7
Overview of security industry trends, technologies, vendor and user strategies	0	0–5	0,4	7	8	0	7
Managerial and policy issues in information security provision	5,7	0,1,3,6	0,2	7	7,8	7	7
When and how policy intervention is needed	–	0,7,8	6,7	7,8	8	0,7,8	7,8

Organizational Education Practices, Assessment Techniques, Experiential & Applied Learning, and Psychology, Behavior, & Human Factors.

- *Intelligence*: includes topics related to Intelligence Fundamentals, Intelligence Frameworks, Intelligence Collection, Intelligence Management, Targeting & Operational Intelligence, Intelligence Production, Intelligence Dissemination, Intelligence Planning, Resource Prioritization, Partner & Organizational Intelligence Integration, and Covert & Specialized Intelligence Techniques.

References

Ramezani, S., Niemi, V., 2024. Cybersecurity education in universities: A comprehensive guide to curriculum development. IEEE Access 12, 61741–61766. doi:10.1109/ACCESS.2024.3392970.

Table 9: Relative importance of CSEC2017 Knowledge Areas for all Work Roles and Work Role Categories contained in the NICE Framework v2.0.0. Knowledge Descriptions contained in each work role were annotated using CurricuLLM and the numbers represent percentage-wise contributions, rounded to the nearest integer.

Work Role	KA0	KA1	KA2	KA3	KA4	KA5	KA6	KA7	KA8
OVERSIGHT and GOVERNANCE (OG)	16	7	3	1	9	3	7	41	12
Communications Security (COMSEC) Management	10	12	5	5	5	4	8	38	11
Cybersecurity Policy and Planning	15	6	2	0	8	2	6	44	17
Cybersecurity Workforce Management	19	4	3	0	7	0	4	51	12
Cybersecurity Curriculum Development	33	6	0	0	11	2	8	29	12
Cybersecurity Instruction	26	8	0	0	17	3	8	26	10
Cybersecurity Legal Advice	17	12	2	0	5	0	12	36	17
Executive Cybersecurity Leadership	8	5	2	0	15	3	5	46	15
Privacy Compliance	10	11	1	1	11	4	13	31	18
Product Support Management	18	4	4	3	7	3	4	48	10
Program Management	16	5	5	3	8	3	5	45	11
Secure Project Management	18	4	4	3	7	3	4	45	10
Security Control Assessment	23	9	6	3	11	5	6	31	7
Systems Authorization	15	13	3	0	8	5	6	39	11
Systems Security Management	14	7	6	4	12	6	5	39	7
Technology Portfolio Management	7	7	2	0	9	0	7	51	16
Technology Program Auditing	16	6	2	0	6	0	6	51	14
DESIGN and DEVELOPMENT (DD)	14	6	10	4	11	9	5	32	8
Cybersecurity Architecture	17	7	9	4	16	13	5	25	5
Enterprise Architecture	14	6	7	5	14	13	6	28	6
Secure Software Development	13	3	19	4	11	11	4	28	9
Secure Systems Development	17	8	12	3	13	8	6	26	7
Software Security Assessment	12	3	19	5	10	9	4	29	9
Systems Requirements Planning	14	9	6	4	8	8	8	34	9
Systems Testing and Evaluation	14	6	9	5	8	9	6	34	10
Technology Research and Development	18	8	8	5	16	7	3	25	10
Operational Technology (OT) Cybersecurity Engineering	5	7	0	0	7	5	5	62	9
IMPLEMENTATION and OPERATION (IO)	20	11	4	2	12	5	8	29	10
Data Analysis	20	13	7	1	7	8	8	25	11
Database Administration	22	17	0	0	7	1	12	29	12
Knowledge Management	31	8	2	0	8	0	8	31	13
Network Operations	20	10	4	4	21	5	7	23	8
Systems Administration	12	8	5	4	21	7	6	28	8
Systems Security Analysis	18	9	6	4	14	8	6	28	7
Technical Support	18	10	1	0	9	4	10	37	10
PROTECTION and DEFENSE (PD)	15	11	3	1	16	5	6	33	9
Defensive Cybersecurity	17	9	6	3	20	8	5	26	6
Digital Forensics	14	23	5	4	11	9	3	22	10
Incident Response	10	10	1	0	19	3	10	37	10
Infrastructure Support	11	11	0	0	22	3	9	34	11
Insider Threat Analysis	15	10	0	0	8	3	5	51	8
Threat Analysis	32	5	2	0	18	3	5	26	8
Vulnerability Analysis	9	10	4	0	14	6	9	36	14
INVESTIGATION (IN)	12	21	3	2	9	7	5	28	12
Cybercrime Investigation	12	18	1	0	9	4	6	35	15
Digital Evidence Analysis	13	25	5	4	10	10	4	21	10